

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1\alpha\beta X}$				
$\mathcal{K}_{0^+}^{\#1\dagger}$	0	0				
$\mathcal{K}_{0^+}^{\#1\dagger\alpha\beta X}$	0	0	$\mathcal{K}_{1^+}^{\#1\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1\alpha}$	$\mathcal{K}_{1^+}^{\#2\alpha}$
	$\mathcal{K}_{1^+}^{\#1\dagger\alpha\beta}$	$-\frac{\gamma_1 k^2}{3}$	$\frac{\gamma_2 k^2}{6\sqrt{2}}$	0	0	
	$\mathcal{K}_{1^+}^{\#2\dagger\alpha\beta}$	$\frac{\gamma_2 k^2}{6\sqrt{2}}$	$\frac{\gamma_3 k^2}{6}$	0	0	
	$\mathcal{K}_{1^+}^{\#1\dagger\alpha}$	0	0	0	0	
	$\mathcal{K}_{1^+}^{\#2\dagger\alpha}$	0	0	0	0	$\mathcal{K}_{2^+}^{\#1\alpha\beta}$
						$\mathcal{K}_{2^+}^{\#1\alpha\beta X}$
					$\mathcal{K}_{2^+}^{\#1\dagger\alpha\beta}$	0
					$\mathcal{K}_{2^+}^{\#1\dagger\alpha\beta X}$	0
						$-k^2 \frac{(4)}{\kappa_1}$

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^-}^{\#1} \alpha\beta\chi$				
$\mathcal{T}_{0^+}^{\#1} \dagger$	0	0				
$\mathcal{T}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{T}_{1^+}^{\#1} \alpha\beta$	$\mathcal{T}_{1^+}^{\#2} \alpha\beta$	$\mathcal{T}_{1^-}^{\#1} \alpha$	$\mathcal{T}_{1^-}^{\#2} \alpha$
$\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{\Upsilon_3 k^2}{6 \text{Det}(1^+)}$	$-\frac{\Upsilon_2 k^2}{6 \sqrt{2} \text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$-\frac{\Upsilon_2 k^2}{6 \sqrt{2} \text{Det}(1^+)}$	$-\frac{\Upsilon_1 k^2}{3 \text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^-}^{\#1} \dagger^\alpha$	0	0	0	0		
$\mathcal{T}_{1^-}^{\#2} \dagger^\alpha$	0	0	0	0	$\mathcal{T}_{2^+}^{\#1} \alpha\beta$	$\mathcal{T}_{2^-}^{\#1} \alpha\beta\chi$
			$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0	
			$\mathcal{T}_{2^-}^{\#1} \dagger^{\alpha\beta\chi}$	0	$-\frac{1}{k^2 \kappa_1^{(4)}}$	

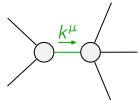
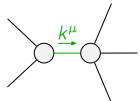
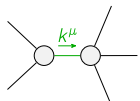
Abbreviations used in matrices

$$Y_1 == 4 \binom{(4)}{K_1} + 3 \binom{(4)}{K_3} \&\& Y_2 == 4 \binom{(4)}{K_1} + 3 \binom{(4)}{K_4} \&\& Y_3 == 5 \binom{(4)}{K_1} + 3 \left(\binom{(4)}{K_3} + \binom{(4)}{K_4} \right) \&\&$$

$$\text{Det}(1^+) = -\frac{1}{24} k^4 (32 \binom{(4)}{k_1}^2 + 3(2 \binom{(4)}{k_3} + \binom{(4)}{k_4})^2 + 12 \binom{(4)}{k_1} (3 \binom{(4)}{k_3} + 2 \binom{(4)}{k_4}))$$

Lagrangian

$$\begin{aligned} & \overset{(4)}{\mathcal{K}}_1 \partial_\beta \mathcal{K}^\delta_{\chi^\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha^\beta} - \overset{(4)}{\mathcal{K}}_1 \partial_\chi \mathcal{K}^\delta_{\beta^\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha^\beta} + \overset{(4)}{\mathcal{K}}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \overset{(4)}{\mathcal{K}}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \\ & 2 \overset{(4)}{\mathcal{K}}_1 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \overset{(4)}{\mathcal{K}}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + 2 \overset{(4)}{\mathcal{K}}_1 \partial^\chi \mathcal{K}^\alpha_{\alpha^\beta} \partial_\delta \mathcal{K}_{\chi\beta}^\delta + \frac{3}{2} \overset{(4)}{\mathcal{K}}_1 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \\ & \frac{1}{2} \overset{(4)}{\mathcal{K}}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{1}{2} \overset{(4)}{\mathcal{K}}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \frac{2}{3} \overset{(4)}{\mathcal{K}}_1 \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - \frac{2}{3} \overset{(4)}{\mathcal{K}}_1 \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{J}_\alpha^{\alpha\beta} == 0$	
$\mathcal{J}_1^{\#2\alpha} == 0$	3	$\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$	
$\mathcal{J}_1^{\#1\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}_\beta^{\beta\chi} + \partial_\chi \partial^\chi \mathcal{J}_\beta^{\beta\alpha} == \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}_\chi^{\chi\delta} == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}_\chi^{\chi\delta} + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	13		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{24 \binom{(4)}{\kappa_1}^2 + 8 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2}{\binom{(4)}{\kappa_1} (32 \binom{(4)}{\kappa_1}^2 + 3 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 + 12 \binom{(4)}{\kappa_1} (3 \binom{(4)}{\kappa_3} + 2 \binom{(4)}{\kappa_4}))}$
	1	0	$-((16 \binom{(4)}{\kappa_1}^2 + 12 \binom{(4)}{\kappa_3}^2 + 12 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} + 3 \binom{(4)}{\kappa_4}^2 + 4 \binom{(4)}{\kappa_1} (5 \binom{(4)}{\kappa_3} + 7 \binom{(4)}{\kappa_4}) +$ $\sqrt{(5520 \binom{(4)}{\kappa_1}^4 + 27 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^4 + 24 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 (31 \binom{(4)}{\kappa_3} + 17 \binom{(4)}{\kappa_4}) + 32 \binom{(4)}{\kappa_1}^3 (367 \binom{(4)}{\kappa_3} + 146 \binom{(4)}{\kappa_4}) +$ $48 \binom{(4)}{\kappa_1}^2 (188 \binom{(4)}{\kappa_3}^2 + 162 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} + 49 \binom{(4)}{\kappa_4}^2))) / (\binom{(4)}{\kappa_1} (32 \binom{(4)}{\kappa_1}^2 + 3 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 + 12 \binom{(4)}{\kappa_1} (3 \binom{(4)}{\kappa_3} + 2 \binom{(4)}{\kappa_4}))))$
	1	0	$(-16 \binom{(4)}{\kappa_1}^2 - 12 \binom{(4)}{\kappa_3}^2 - 12 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} - 3 \binom{(4)}{\kappa_4}^2 - 4 \binom{(4)}{\kappa_1} (5 \binom{(4)}{\kappa_3} + 7 \binom{(4)}{\kappa_4}) +$ $\sqrt{(5520 \binom{(4)}{\kappa_1}^4 + 27 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^4 + 24 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 (31 \binom{(4)}{\kappa_3} + 17 \binom{(4)}{\kappa_4}) +$ $32 \binom{(4)}{\kappa_1}^3 (367 \binom{(4)}{\kappa_3} + 146 \binom{(4)}{\kappa_4}) + 48 \binom{(4)}{\kappa_1}^2 (188 \binom{(4)}{\kappa_3}^2 + 162 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} + 49 \binom{(4)}{\kappa_4}^2))) /$ $(\binom{(4)}{\kappa_1} (32 \binom{(4)}{\kappa_1}^2 + 3 (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 + 12 \binom{(4)}{\kappa_1} (3 \binom{(4)}{\kappa_3} + 2 \binom{(4)}{\kappa_4}))))$
Resolved unitarity condition(s):	(Demonstrably impossible)		